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Integration of tear fluid biomarkers and machine learning for the early detection of keratoconus

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Abstract

Background Keratoconus (KC) is a progressive corneal disorder that leads to visual impairment owing to corneal thinning and deformation. While current diagnostic methods are effective in the later stages, early detection of KC remains challenging. This study aimed to investigate the potential of tear fluid biomarkers combined with machine learning (ML) models for the early diagnosis of KC.

Methods A total of 370 participants were recruited, including 134 patients with KC, 93 with early KC, and 143 normal controls. Cytokine levels in tear fluid were measured using multiplex cytokine analysis. Clinical parameters were evaluated, and nine ML models were employed to predict early KC risk based on tear biomarkers and clinical data.

Results Significant increases in IL-1 β , Galectin-1, and Galectin-3 levels were observed in the KC and early KC groups compared to controls ($p < 0.05$ for all). The Random Forest model demonstrated the highest accuracy when integrating tear biomarker levels and clinical parameters. Key biomarkers, such as Galectin-3, IL-1 β , and Galectin-1, in combination with clinical parameters (D-index, BE, and Kmax), significantly enhanced prediction accuracy. Furthermore, a web application was developed to facilitate clinical deployment, enabling personalized risk assessment for early KC.

Conclusions The study highlight the potential of integrating noninvasive tear fluid biomarkers, Pentacam parameters and ML techniques to improve the early detection of KC.

Keywords Keratoconus, Machine learning, Random forest model, Tear biomarkers, Multiplex detection, Diagnosis

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Introduction

Keratoconus (KC) is a progressive corneal disorder characterized by thinning and cone-shaped deformation of the cornea, leading to irregular astigmatism and severe visual impairment [1]. Recent research has advanced our understanding of KC pathogenesis, suggesting that genetic factors, environmental influences, and inflammatory processes collectively contribute to disease progression [2–4]. Despite these insights, the mechanisms underlying KC progression remain complex and incompletely understood, which limits the development of effective early diagnostic tools.

Currently, KC diagnosis largely depends on corneal topography and tomography, which detect irregularities in corneal shape and thickness. However, these techniques are often effective only after structural changes occur, making them less useful for identifying the early stages of KC. Other modalities, such as corneal biomechanical measurements, provide additional insights, but lack sufficient specificity and sensitivity for early KC screening [5]. This diagnostic limitation underscores the need for innovative approaches capable of detecting KC in the subclinical or early stages to enable timely intervention.

Emerging evidence suggests that tear fluid, a readily accessible source of biomarkers, may play a critical role in the diagnosis of KC [6]. The ocular surface environment, particularly the tear film, contains numerous proteins, cytokines, and other molecules that affect ocular health and the inflammatory status [7–9]. Recent studies have identified various biomarkers for KC tears [10–12]. The levels of inflammatory cytokines are elevated in KC tears, indicating that inflammation may contribute to corneal weakening and ectasia [13–15]. Galectin-1 and Galectin-3, which are involved in immune regulation, exhibit increased expression in KC and are associated with corneal matrix remodeling, suggesting their involvement in disease pathogenesis [16] and underscoring their diagnostic potential.

Machine learning (ML) has recently shown promise for detecting KC by analyzing complex patterns in biomedical data [17]. Studies have indicated that ML models can improve diagnostic accuracy by identifying subtle biomarker interactions that may be missed using conventional statistical methods [18, 19]. Combining data from corneal topography with ML has enhanced early KC detection, outperforming the traditional diagnostic criteria [20, 21]. Moreover, integrating tear biomarker profiles with logistic regression can predict KC progression [22]. These findings suggest that ML-based models incorporating tear biomarkers may be highly effective in predicting KC onset before structural changes manifest. However, related studies remain limited.

In this study, we integrated multiplex immuno-bead detection of tear fluid biomarkers with ML algorithms to investigate their potential for early diagnosis of KC. Through the identification of biomarker profiles associated with KC progression and the development of a predictive ML model, we aimed to enhance the early identification and diagnostic accuracy of KC and to facilitate timely intervention.

Subjects and methods

Participant enrollment and grouping

Patients diagnosed with KC or early KC at the Ophthalmic Center of the First Affiliated Hospital of Fujian Medical University between July 2022 and December 2023 were enrolled in this study.

Two senior ophthalmologists (L.H. and Y.L.) independently evaluated and classified each sample. The KC group consisted of patients with bilateral KC who met the following criteria: a history of myopia and astigmatism, best corrected visual acuity (BCVA) below 16/20, at least one positive slit-lamp biomicroscope sign of KC, and abnormal corneal tomography findings, such as an inferior-superior asymmetry value exceeding 3.0 diopters (D) within a 3 mm radius on the anterior corneal surface, or an inter-eye difference in the central anterior curvature of more than 1.0 D [23]. Individuals with KC features in one eye, whereas the other eye presented normal findings on slit lamp, retinoscopy, and topographic evaluations. The fellow eyes that showed no asymmetrical bowtie pattern, focal steepening, or inferior steepening and maintained a BCVA of 16/20 or better were included in the early KC group [24]. The normal control group comprised individuals with BCVA of 16/20 or better and no ocular abnormalities beyond myopia. For each participant in the normal control group, data from one eye were randomly selected using a computer-generated random number method to ensure that the assumption of independence was satisfied. The participants were advised to discontinue wearing soft or hard contact lenses for at least 2 or 4 weeks, respectively, before the examination. None of the participants reported a history of systemic disease, head trauma, neurological disorders, or ophthalmic disease other than refractive error or primary KC, nor had they undergone any ophthalmic surgery.

The study was approved by the Ethics Committee of the First Affiliated Hospital of Fujian Medical University (MRTCA, ECFMH of FMU [2022] 323-2) and performed in strict adherence to the tenets of the Declaration of Helsinki. Informed consent forms were obtained from the patients or their guardians. The flowchart of this study is shown in Fig. 1.

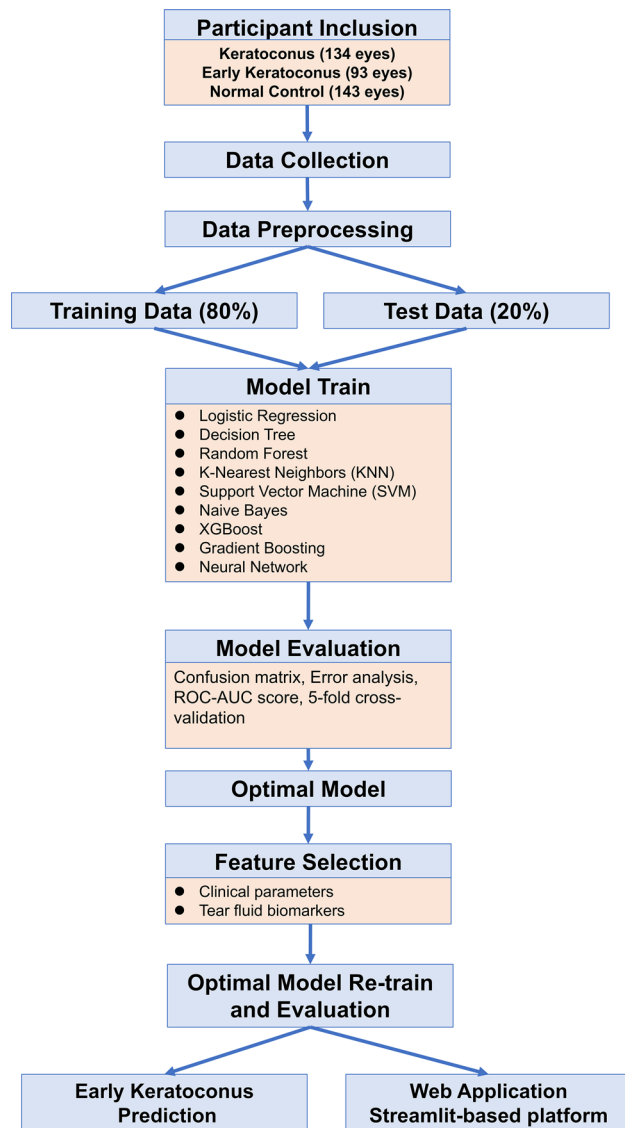


Fig. 1 Flowchart of this study

Data collection

Rotating Scheimpflug tomography (Pentacam HR, Oculus Optikgeräte GmbH, Wetzlar, Germany) was performed by an experienced ophthalmic technician (X.M.), with only those results labeled “QS” (quality control) as “OK” selected for analysis. The following parameters were recorded: K1 (D), K2 (D), Kmax (D), Kmax Front (D), Astigmatism (D), Pachy Min (μm), FE, BE, ARTave, ARTmax, and D-index. Axial length (AL) was measured using optical biometry (Lenstar LS900, Haag-Streit AG, Koeniz, Switzerland). Cycloplegic refraction was performed with 0.5% tropicamide eye drops. The refractive values were converted to spherical equivalents (SE), calculated as the algebraic sum of the sphere and half of the cylinder. Additionally, the following clinical characteristics and examinations were documented: age, sex,

uncorrected visual acuity (UCVA), best corrected visual acuity (BCVA), noncontact intraocular pressure (NCT), corneal tomography, slit-lamp examination, and fundus assessment. UCVA and BCVA were recorded as Log-MAR visual acuity. All measurements were repeated three times for each eye.

Tear collection

Tear samples were collected by the same examiner within the same time window, between 9:00 AM and 11:00 AM, prior to any other ocular examinations, and from unilateral eyes. The procedure was performed as described previously [25], entailed the administration of 20 μL of sterile saline solution into the conjunctival fornix. Participants were instructed to gently close their eyes and rotate them twice during this period. After 20 s, tear samples were collected without anesthesia using calibrated 10-μL glass microcapillary tubes (Drummond Microcaps, Merck, USA) from the temporal inferior tear meniscus to minimize ocular surface irritation. The samples were labeled, placed in Eppendorf tubes, and stored at -80 °C until analysis.

Tear analysis

Tear samples were diluted 1:20 in phosphate-buffered saline to a final volume of 100 μL according to the protocols provided by the multiplex immuno-bead assay kit (Luminex Human® Discovery Assay, LXSAM-30, R&D, USA), allowing for the simultaneous analysis of 30 biomarkers. The cytokine concentrations were measured using a Luminex 200 system (DiaSorin, USA).

Model construction and evaluation

In this study, nine ML models were trained to predict the risks of KC and early KC. The dataset was divided into training (80%) and testing (20%) sets to evaluate the model performance on unseen data. The models included Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes, XGBoost, Gradient Boosting, and Neural Network. Model performance was evaluated based on accuracy, sensitivity, precision, F1 score, Kappa score, and the Area Under the Curve (AUC) for both the training and testing datasets.

The most predictive model was further refined using feature selection techniques, isolating key features that contributed significantly to the risk of early KC. The variables were categorized into three groups: tear fluid biomarkers, clinical parameters, and all the variables combined. Three separate models were constructed for each dataset (tear fluid biomarkers, clinical parameters, and all variables) using both the training and testing sets. The model performances, including accuracy, sensitivity, precision, F1 score, and Kappa score, were compared

across the three datasets. The effects of incorporating tear fluid biomarkers, clinical parameters, and both on the model performance were analyzed. The random forest model was retrained using the top three ranked features of the two clusters (Pentacam parameters and tear biomarkers), based on importance scores. To assess the robustness of the optimal model, a confusion matrix, classification report, and 5-fold cross-validation procedure were used. In addition, a receiver operating characteristic (ROC) curve was generated for each class.

Network calculator

To enhance the clinical utility, the final prediction model was implemented as a web-based platform via StreamLit. The application calculates and displays the probability of early KC by inputting the values of the selected features.

Statistical analysis

Statistical analysis was conducted using PyCharm (Edition 2020.1.2 × 64) integrated with Python (Version 3.11) on a Windows 11 operating system. Continuous variables were recorded as the mean and standard deviation (SD), whereas categorical variables were recorded as numbers. The consistency of variables with a normal distribution was investigated via both visual (histogram and probability graphics) and analytical methods (Kolmogorov–Smirnov test). The *chi-square* test was used to analyze the categorical data. The Kruskal–Wallis test was applied to

nonnormally distributed variables. If the overall test was significant, pairwise comparisons between groups were performed using the Mann–Whitney U test with Bonferroni correction for multiple comparisons. *P* values of < 0.05 were considered statistically significant.

Results

Baseline characteristics of the participants

The study included 134 participants in the KC group, 93 in the early KC group, and 143 in the normal control group. A summary of the demographic and clinical characteristics of the study population is presented in Table 1 and Table S1.

Multiplex cytokine analysis of tear fluid in all the participants

All 30 biomarkers were successfully measured in all samples. The cytokine levels in the tear fluid of all participants are detailed in Table 2. Significant differences in the levels of cytokines (IL-1 β , IL-2, IL-3, IL-6, IL-10, IL-17/IL-17A, TNF- α , Galectin-1, Galectin-3, Galectin-9, MMP-1, MMP-2, MMP-13, and GM-CSF) were detected among the early KC, KC, and normal control groups (all *P* < 0.05). Moreover, the levels of cytokines (IL-1 β , Galectin-1, and Galectin-3) in KC and early KC were greater than those in the normal control (all *P* < 0.05, Fig. 2). The levels of Galectin-1 and Galectin-3 in KC were elevated compared to those in early KC (all *P* < 0.05). Conversely,

Table 1 Baseline characteristics of the participants

Group	Normal <i>n</i> = 143	KC <i>N</i> = 134	Early KC <i>N</i> = 93	<i>P</i> value
sex (Male/Female)	81/62	97/37	60/33	0.024 [#]
age (years)	24.29 ± 6.26	23.26 ± 8.12	21.89 ± 7.39	0.001*
NCT (mmHg)	16.40 ± 2.81	12.37 ± 3.04	15.73 ± 2.80	0.000*
UCVA (LogMAR)	1.13 ± 0.40	1.09 ± 0.60	1.11 ± 0.51	0.776*
BCVA (LogMAR)	0.00 ± 0.04	0.50 ± 0.57	0.04 ± 0.22	0.000*
SE (D)	-5.02 ± 2.23	-7.76 ± 3.87	-4.91 ± 2.78	0.000*
K1 (D)	42.23 ± 1.10	47.39 ± 7.96	42.10 ± 1.35	0.000*
K2 (D)	43.71 ± 1.48	52.08 ± 9.34	47.95 ± 40.92	0.000*
Kmax (D)	44.32 ± 1.55	58.23 ± 10.46	44.72 ± 2.21	0.000*
Kmax Front (D)	44.33 ± 1.55	57.97 ± 12.32	44.40 ± 1.93	0.000*
Astig (D)	1.49 ± 0.87	4.67 ± 6.13	1.58 ± 0.83	0.000*
Pachy Min (μm)	537.96 ± 29.00	473.28 ± 47.74	532.61 ± 35.79	0.000*
FE	2.06 ± 1.38	18.94 ± 23.64	3.32 ± 4.20	0.000*
BE	4.17 ± 2.48	48.56 ± 33.52	8.39 ± 8.83	0.000*
ARTave	1.02 ± 0.12	2.19 ± 1.04	1.21 ± 0.79	0.000*
ARTmax	438.10 ± 70.20	177.75 ± 93.22	393.71 ± 112.14	0.000*
D-index	0.92 ± 0.55	9.21 ± 6.94	1.69 ± 2.18	0.000*
AL (mm)	27.57 ± 20.78	25.40 ± 1.54	25.83 ± 1.30	0.007*

[#]*Chi-square* test and *Kruskal–Wallis test among the KC, early KC and normal control groups

KC: keratoconus; NCT: noncontact intraocular pressure; UCVA: uncorrected visual acuity; BCVA: best corrected visual acuity; SE: spherical equivalent; D: diopters; LogMAR: logarithm of the minimum angle of resolution; K1: flat keratometry of the corneal anterior surface; K2: steep keratometry of the corneal anterior surface; Kmax: maximum axial curvature; Kmax Front: maximum axial curvature of the front surface; Astig: astigmatism; Pachy Min: minimum corneal thickness; FE: front elevation; BE: back elevation; ARTave: average Ambrosio's relational thickness; ARTmax: maximum Ambrosio's relational thickness; D-index: Belin/Ambrósio deviation index; AL: axial length

Table 2 Cytokine levels in the tear fluid of all participants

Group	Normal (pg/ml)	KC (pg/ml)	Early KC (pg/ml)	P value
IL-1 β	12.44 $\pm$ 11.91	41.97 $\pm$ 103.69	61.85 $\pm$ 98.21	0.000*
IL-2	19.06 $\pm$ 12.04	15.13 $\pm$ 8.54	19.60 $\pm$ 12.74	0.004*
IL-3	31.69 $\pm$ 16.32	29.25 $\pm$ 13.04	28.43 $\pm$ 17.10	0.028*
IL-4	99.84 $\pm$ 66.11	92.63 $\pm$ 58.82	96.81 $\pm$ 67.81	0.704*
IL-5	3.38 $\pm$ 2.31	3.22 $\pm$ 2.57	3.67 $\pm$ 3.38	0.208*
IL-6	4.11 $\pm$ 3.87	6.03 $\pm$ 4.97	5.82 $\pm$ 6.59	0.000*
IL-8/CXCL8	248.77 $\pm$ 589.30	225.52 $\pm$ 180.51	564.38 $\pm$ 1407.91	0.127*
IL-10	6.90 $\pm$ 7.65	8.13 $\pm$ 6.21	9.46 $\pm$ 10.15	0.046*
IL-12p70	55.74 $\pm$ 48.67	65.83 $\pm$ 51.62	58.41 $\pm$ 52.69	0.116*
IL-13	436.23 $\pm$ 262.26	431.81 $\pm$ 293.77	414.30 $\pm$ 255.46	0.645*
IL-17/IL-17A	25.86 $\pm$ 31.83	34.06 $\pm$ 39.11	39.13 $\pm$ 40.15	0.019*
TNF- α	21.95 $\pm$ 30.22	27.55 $\pm$ 30.46	30.15 $\pm$ 37.10	0.023*
IFN- γ	17.19 $\pm$ 9.78	19.84 $\pm$ 15.02	19.23 $\pm$ 12.46	0.220*
CCL2/JE/MCP-1	125.84 $\pm$ 150.07	95.80 $\pm$ 135.29	111.17 $\pm$ 154.22	0.307*
CCL4/MIP-1 beta	386.56 $\pm$ 245.77	374.65 $\pm$ 236.90	388.31 $\pm$ 258.74	0.950*
CCL5/RANTES	42.78 $\pm$ 36.30	38.69 $\pm$ 32.67	49.12 $\pm$ 45.73	0.665*
Coagulation Factor XIV/Protein C	708.00 $\pm$ 672.82	496.01 $\pm$ 360.00	672.67 $\pm$ 606.87	0.799*
RAGE/AGER	692.22 $\pm$ 610.03	661.71 $\pm$ 680.80	694.99 $\pm$ 656.11	0.201*
FGF basic/FGF2/bFGF	10.05 $\pm$ 9.79	8.00 $\pm$ 5.98	9.36 $\pm$ 8.61	0.263*
CX3CL1/Fractalkine	3270.51 $\pm$ 1852.26	3443.68 $\pm$ 2213.81	3710.16 $\pm$ 2453.89	0.545*
Galectin-1	2593.74 $\pm$ 2623.83	9383.22 $\pm$ 6324.54	5673.68 $\pm$ 4826.94	0.000*
Galectin-3	7160.22 $\pm$ 3971.11	10665.87 $\pm$ 4555.07	9206.52 $\pm$ 3978.47	0.000*
Galectin-9	23783.67 $\pm$ 19997.57	28947.97 $\pm$ 18710.94	26601.28 $\pm$ 19007.64	0.006*
CA15-3/MUC-1	1.91 $\pm$ 3.03	2.92 $\pm$ 11.68	9.34 $\pm$ 26.41	0.085*
CA125/MUC16	1560.30 $\pm$ 927.70	1445.21 $\pm$ 752.08	1500.19 $\pm$ 979.29	0.634*
MMP-1	15.59 $\pm$ 10.33	20.41 $\pm$ 12.50	19.42 $\pm$ 22.23	0.000*
MMP-2	2651.76 $\pm$ 1401.93	3409.10 $\pm$ 962.07	2547.30 $\pm$ 1163.31	0.000*
MMP-9	7171.80 $\pm$ 9208.82	9999.41 $\pm$ 11297.30	10044.81 $\pm$ 13726.25	0.212*
MMP-13	344.32 $\pm$ 231.05	460.41 $\pm$ 154.82	350.85 $\pm$ 203.43	0.000*
GM-CSF	8.15 $\pm$ 5.61	6.37 $\pm$ 4.49	7.51 $\pm$ 5.08	0.004*

*Kruskal–Wallis test among the KC, early KC and normal control groups

no significant difference was detected in the IL-1 β level between the KC and early KC groups ($P < 0.05$).

Random forest model exhibits superior predictive performance

The random forest model trained with all the parameter data, consistently outperformed the other models in both the training and testing sets, with higher values for ‘Accuracy’, ‘Sensitivity (Recall)’, ‘Precision’, ‘F1 score’, and ‘Kappa score’ (Fig. 3a + b). Additionally, when cross-validation was used to calculate the AUC for each model, the random forest model exhibited the highest AUC value compared to the other models (Fig. 3c).

Random forest model with all parameter data outperforms other models

Compared with models trained using only tear fluid biomarkers or clinical parameter data, the random forest model, which incorporated all parameter data, demonstrated superior predictive performance. In both the training and testing sets, this model yielded higher

values for ‘Accuracy’, ‘Sensitivity (Recall)’, ‘Precision’, ‘F1 score’, and ‘Kappa score’ than the other two models did (Fig. 4a + b).

Performance of the random forest model in predicting early KC

The random forest model trained with all the parameters achieved a 5-fold cross-validation score of 0.868 (Fig. 5a), with an ROC curve AUC value of 0.939 for predicting early KC (Fig. 5b). Among the clinical parameters, the D-index, BE, and Kmax demonstrated the highest feature importance scores (Fig. 5c). Among the tear fluid cytokines, Galectin-3, IL-1 β , and Galectin-1 showed higher feature importance scores (Fig. 5c).

Using the six selected features (D-index, BE, Kmax, Galectin-3, IL-1 β , and Galectin-1), a new random forest model was trained to improve early KC prediction. Error analysis revealed a mean prediction error (MPE) of 0.041, standard deviation (SD) of 0.625, mean squared error (MSE) of 0.391, mean absolute error (MAE) of 0.203, and median absolute error (MedAE) of 0.000. The AUC

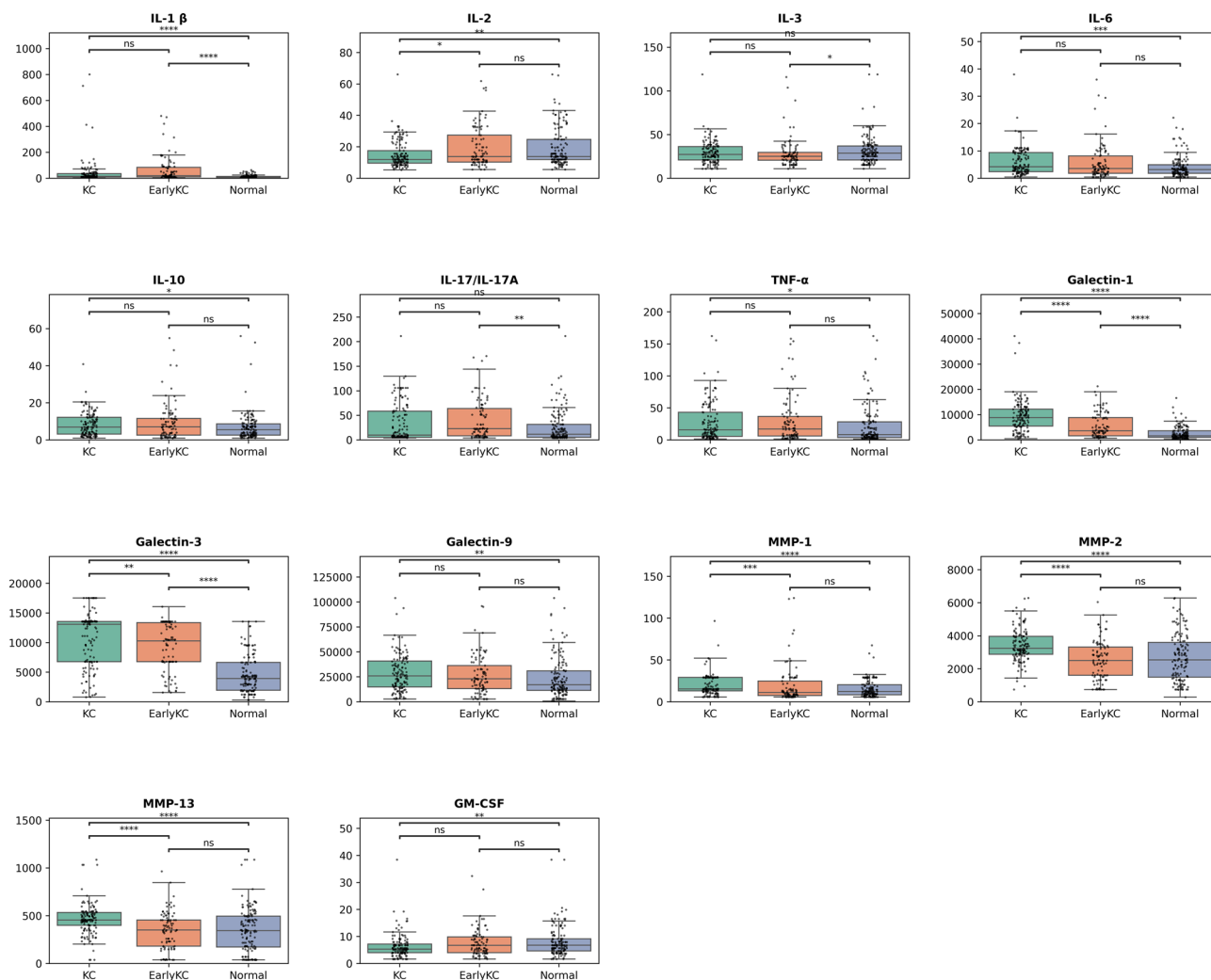


Fig. 2 Boxplots of Tear Cytokines with Significant Intergroup Differences. Boxplots illustrate the distribution of tear cytokine levels among normal controls, early KC, and KC groups. Only cytokines showing significant overall differences by the Kruskal–Wallis test ($p < 0.05$) are presented. Pairwise comparisons were performed using the Mann–Whitney U test with Bonferroni correction. Statistical significance is indicated as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, and *ns* = not significant. Horizontal lines denote the group comparisons, and data points represent individual subjects

for predicting early KC was 0.955 (normal, 0.976; KC, 1.000), surpassing that of the model trained with all the parameters (Fig. 5d). The confusion matrix demonstrated excellent model performance (Fig. 5e), with near-perfect classification for KC (33/34 correctly classified), high accuracy for normal eyes (21/25 correctly classified), and reliable performance in identifying early KC (12/15 correctly classified).

Deployment of the web calculator for the early KC

The final prediction model was integrated into a web application for clinical use, as illustrated in Fig. 6. By entering the actual values of the six features required by the model, the application can automatically assess the risk of early KC in individual patients. The web application is accessible online via the following link: <https://qppzfzj3wtjrkqcwv7n4bm.streamlit.app/>.

Discussion

This study aimed to explore the diagnostic potential of tear fluid biomarkers and clinical parameters for early KC and to evaluate the accuracy of ML models in predicting early KC. We detected elevated levels of cytokines (IL-1 β , Galectin-1, and Galectin-3) in the tear fluid of patients with KC and early KC, suggesting their role as potential biomarkers for this disease. Among the nine ML models evaluated, the random forest model demonstrated superior predictive performance, particularly when trained with both tear fluid biomarkers and clinical parameters. The random forest model, retrained using Pentacam parameters (D-index, BE, and Kmax) and tear biomarkers (Galectin-1, Galectin-3, and IL-1 β), the top three ranked features of the two clusters based on importance scores,

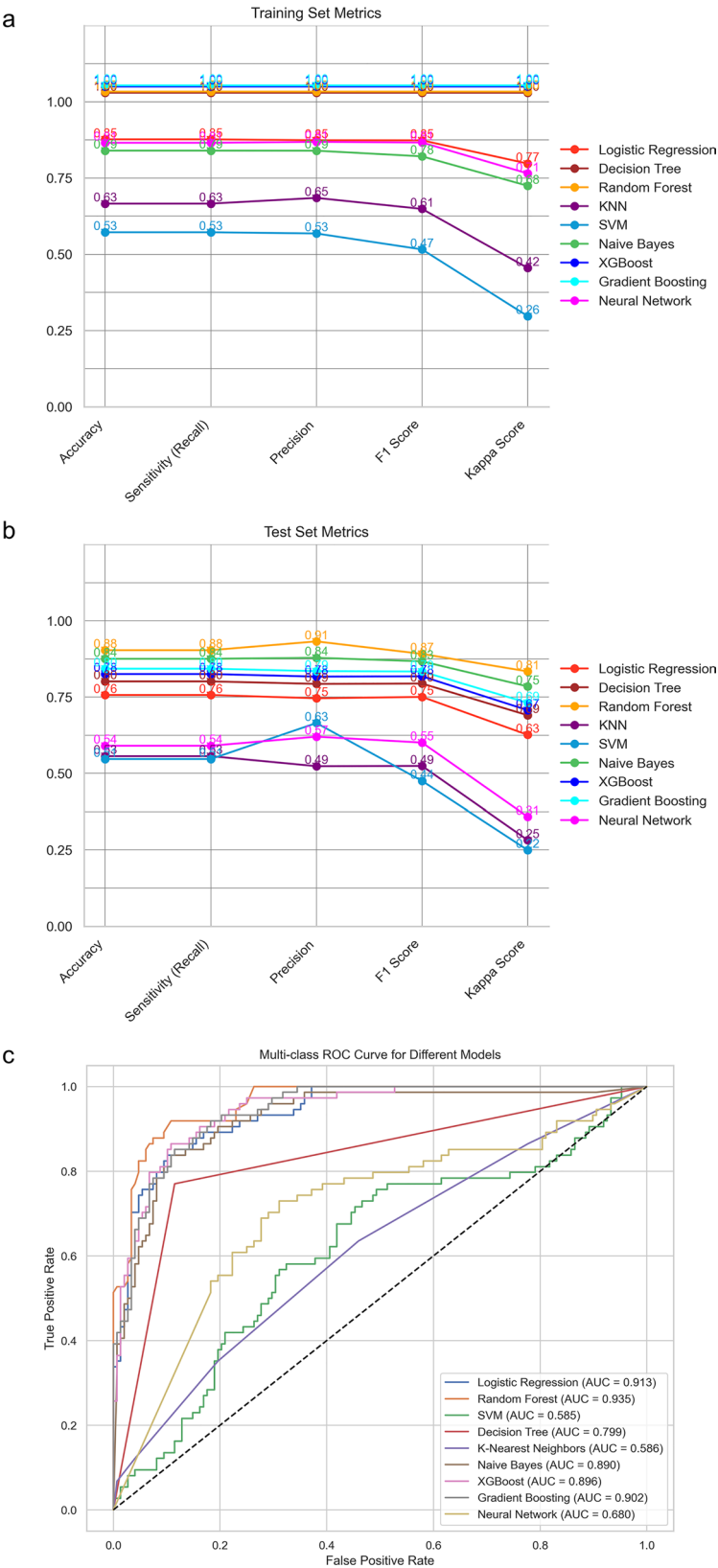


Fig. 3 Performance of machine learning (ML) in the training and test sets. The Random Forest model trained with all the parameter data, demonstrated superior performance compared to the other models on both training (**a**) and test (**b**) datasets, achieving higher values for accuracy, sensitivity (recall), precision, F1 score, and Kappa score. Cross-validation analysis further confirmed its effectiveness, with the random forest model yielding the highest AUC among all the evaluated models (**c**)

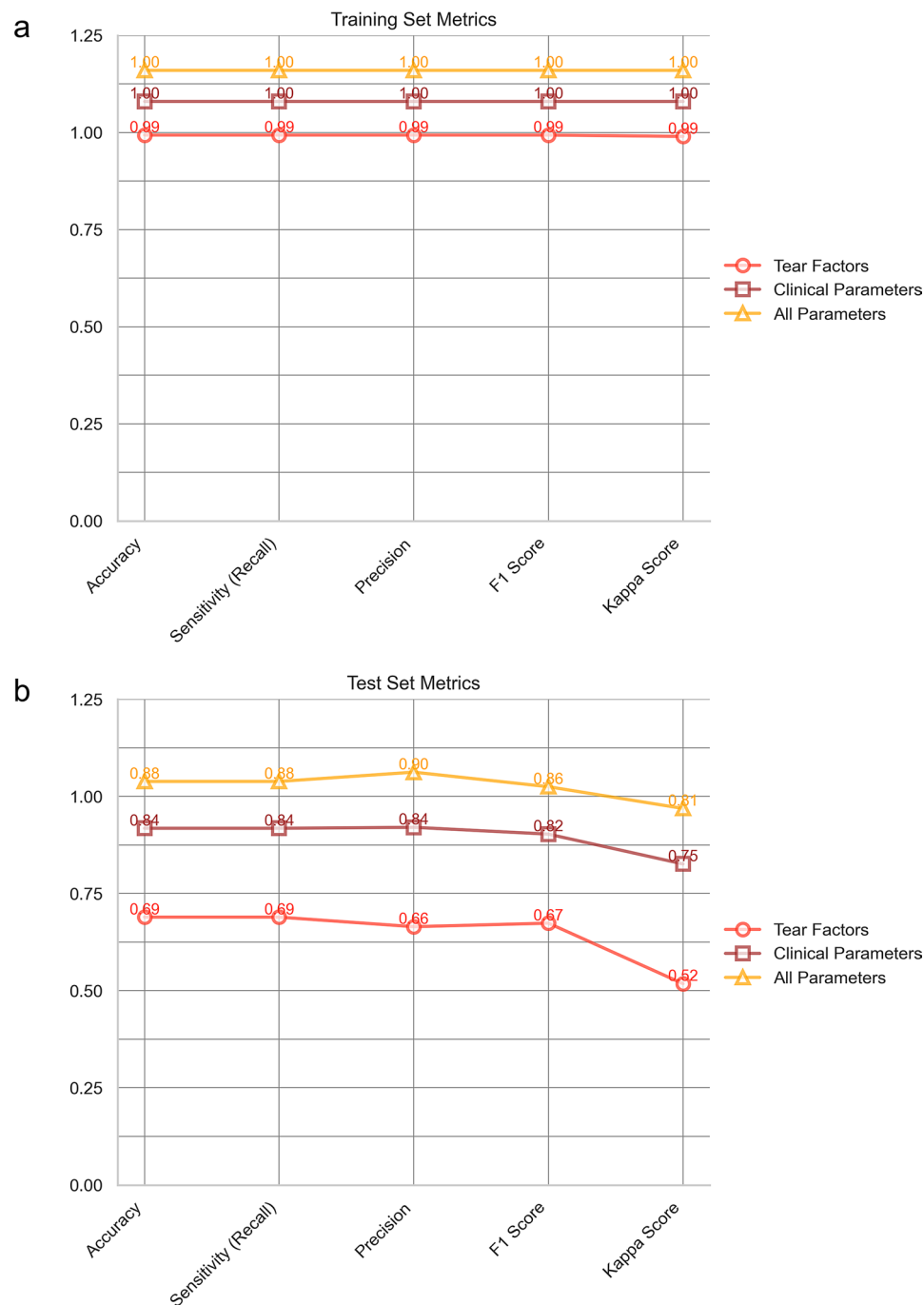


Fig. 4 Comparative Performance of Random Forest Models with Different Parameter Sets. Compared with models trained solely on tear fluid biomarkers or clinical parameters, the random forest model, which incorporated all parameters, exhibited superior predictive performance. On both the training (**a**) and testing (**b**) datasets, the comprehensive model achieved higher accuracy, sensitivity (recall), precision, F1 score, and Kappa score than the models that utilized only subsets of data

demonstrated optimal performance in predicting early KC.

Our findings align with those of previous studies that highlighted the role of tear fluid biomarkers in KC diagnosis [26]. Inflammatory cytokines in tears, such as IL-1 β , which were significantly elevated in our study, have been

implicated in the pathophysiology of KC, likely by contributing to keratocyte apoptosis, tissue degradation, and remodeling [7, 11]. These findings corroborate earlier work that identified proinflammatory cytokines as closely associated with the progression and severity of KC [27, 28]. Although IL-1 β level appeared slightly higher in early

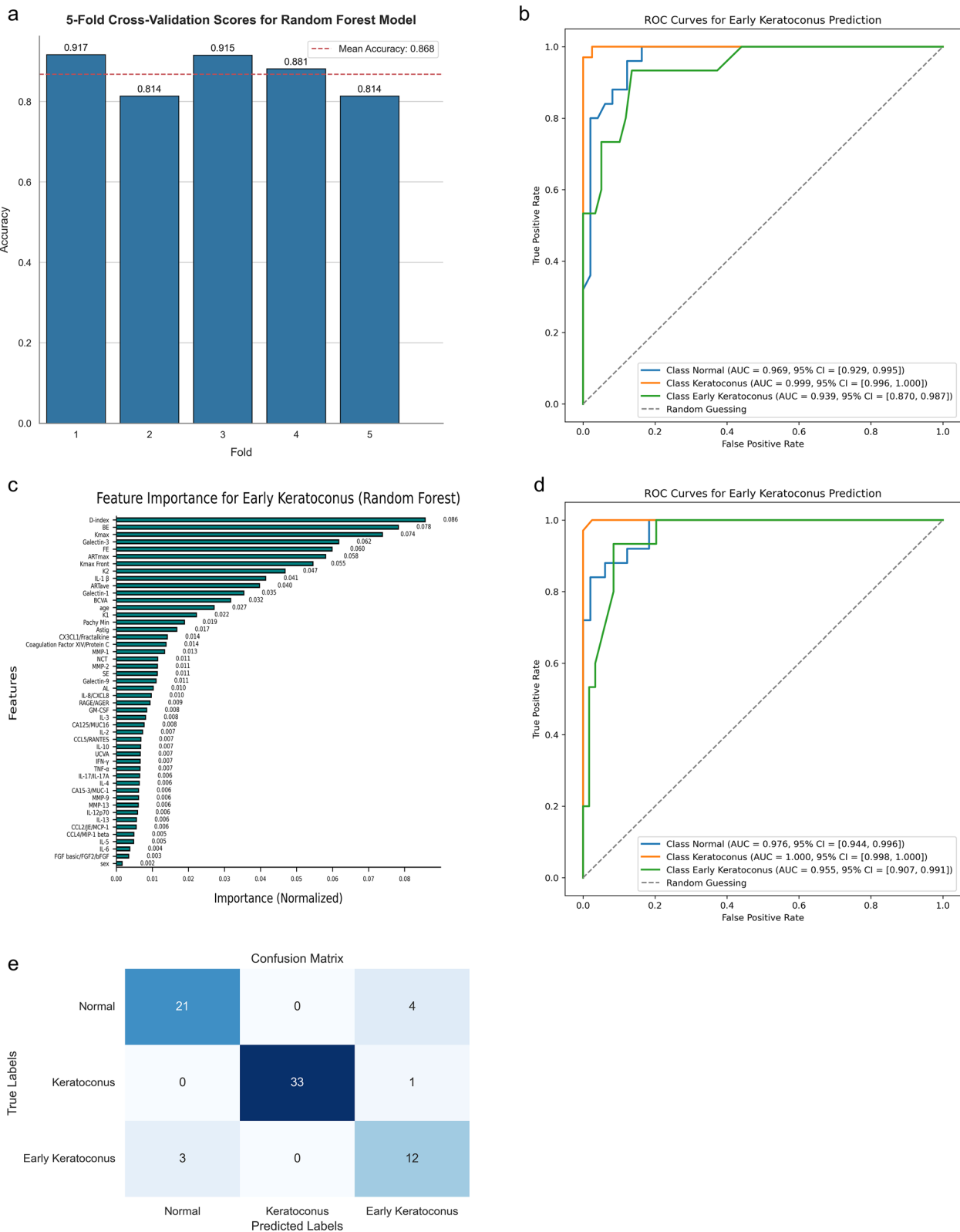


Fig. 5 Feature Selection and Model Optimization for Early KC Prediction. The random forest model trained with all the parameters achieved a 5-fold cross-validation score of 0.868 **(a)** and an ROC curve AUC of 0.939 for predicting early KC **(b)**. Feature importance analysis revealed the D-index, BE, and Kmax as key clinical parameters, whereas Galectin-3, IL-1β, and Galectin-1 were identified as critical tear fluid cytokines **(c)**. A new random forest model was trained using these six selected features. This model achieved an AUC of 0.955 (normal: 0.976, KC: 1.000), outperforming the model trained with all the parameters **(d)**. The confusion matrix for the optimized model demonstrated its excellent predictive accuracy, particularly in identifying KC and normal cases **(e)**

Risk Prediction of Early Keratoconus

Please enter the following eigenvalues:

D-index:	2.58	-	+
BE:	10.00	-	+
Kmax(D):	45.90	-	+
Galectin-1(pg/mL):	13500.00	-	+
Galectin-3(pg/mL):	10000.00	-	+
IL-1 beta(pg/mL):	3000.00	-	+
Predict			

early_keratoconus_risk: 90.00%

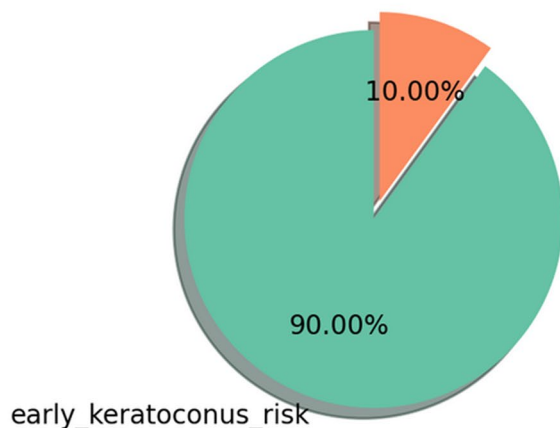


Fig. 6 Deployment of the Web-Based Prediction Tool for Early KC. The final prediction model was implemented in a web application for clinical use. By inputting the actual values of the six selected features (D-index, BE, Kmax, Galectin-3, IL-1 β , and Galectin-1), the application automatically assesses the risk of early KC in individual patients

KC than in KC, this difference was not statistically significant. Its consistent elevation compared with that of the controls and strong predictive contribution in the model may support its utility as an early diagnostic rather than a progression-related biomarker.

Moreover, our study highlighted elevated levels of Galectin-1 and Galectin-3 in tear fluid, which exhibited high feature importance scores in the early KC prediction model, thus underscoring their potential as diagnostic biomarkers. Galectins are a family of lectins characterized by their specific binding affinity for β -galactosidase carbohydrates and a conserved consensus amino acid

sequence within the carbohydrate recognition domain (CRD). Galectin-1 and Galectin-3, which are both involved in immune regulation and extracellular matrix remodeling, were elevated in KC patients, supporting their role in modulating the balance between anti-inflammatory and proinflammatory responses in the cornea and implicating their involvement in disease pathogenesis [16]. As the first galectin identified by its hemagglutination properties, Galectin-1 acts as an effector by modulating innate and adaptive immune responses [29, 30]. Galectin-1 is constitutively expressed at low levels in normal corneal epithelial and stromal cell cytoplasm, whereas its levels in tears are relatively high even in control [31], which may account for the measurable baseline concentration of Galectin-1 detected in control samples. Elevated levels in KC have also been reported in previous studies [16], suggesting that its upregulation reflects inflammatory and remodeling processes in corneal pathology. Similarly, Galectin-3, the only chimera-type galectin [32], is naturally abundant in tears due to its role in maintaining the ocular surface glycocalyx and its ability to form pentamers through interactions with transmembrane mucin glycans [33]. Its further elevation in KC may be associated with matrix degradation and proinflammatory amplification pathways [34, 35]. Supporting this, exogenous Galectin-3 has been shown to bind CD147 to induce MMP-9 expression, enhance IL-1 β -mediated inflammation in corneal epithelial cells, and modulate inflammatory responses in corneal fibroblasts [35, 36]. These findings highlight the dual roles of Galectin-1 and Galectin-3 as physiological tear components and as biomarkers of corneal disease activity.

Our findings regarding Galectin-1, Galectin-3, and IL-1 β are consistent with previous studies; however, other tear biomarkers, including MMP-9, IL-6, and RAGE, did not exhibit significant differences. Previous studies have reported increased concentrations of MMP-9, IL-6, and RAGE in the tear fluid of patients [37–39], and this discrepancy may be attributable to differences in disease stage, assay sensitivity, or cohort demographics [40]. Our negative finding underscores the heterogeneity of KC biomarker expression and highlights the need for further longitudinal studies.

Furthermore, the integration of ML models with biomarker data in our study demonstrated the potential of this approach to improve KC diagnosis. ML techniques have been successfully applied to other ocular surface diseases, with promising results in predicting disease outcomes and facilitating early diagnosis. Our study extends this application to KC, showing that the combination of clinical parameters and tear fluid biomarkers significantly improves the predictive accuracy compared with traditional diagnostic methods. One of the key strengths of this study lies in its noninvasive approach, which utilizes

tear fluid biomarkers to predict early KC. The use of multiplex immunoassays enables the simultaneous analysis of multiple biomarkers, providing a comprehensive profile of tear fluid composition. Moreover, the application of advanced ML algorithms, particularly the random forest model, demonstrated robust predictive capabilities for early KC detection.

However, this study has several limitations. First, the sample size, while adequate, was relatively small, and the study population was limited to a specific cohort from a single institution. Further external validation with a larger and more diverse population is necessary to confirm the generalizability of our findings. Additionally, the cross-sectional nature of the study limits our ability to track biomarker changes over time and to assess their predictive value for disease progression. Longitudinal studies would be beneficial in establishing the temporal relationship between tear biomarker levels and KC progression. Lastly, while we focused on specific biomarkers, the tear film is a complex milieu, and additional proteins or molecular pathways, which were not addressed in this study, may play important roles in KC.

In conclusion, our study highlights the potential of tear fluid biomarkers, particularly Galectin-1, Galectin-3, and IL-1 β , as noninvasive diagnostic tools for early KC detection. Integrating these biomarkers with the Pentacam parameters (D-index, BE, and Kmax) in ML models, particularly the random forest model, offers a promising strategy for improving early KC diagnosis, outperforming traditional diagnostic methods. This integrated approach could pave the way for personalized screening strategies and earlier interventions, potentially improving the outcomes of patients with KC.

Abbreviations

KC	Keratoconus
NCT	Noncontact intraocular pressure
UCVA	Uncorrected visual acuity
BCVA	Best corrected visual acuity
SE	Spherical equivalent
D	Diopters
LogMAR	Logarithm of the minimum angle of resolution
K1	Flat keratometry of the corneal anterior surface
K2	Steep keratometry of the corneal anterior surface
Kmax	Maximum axial curvature
Kmax Front	maximum axial curvature of the front surface
Astig	Astigmatism
Pachy Min	Minimum corneal thickness
FE	Front elevation
BE	Back elevation
ARTave	Average Ambrosio's relational thickness
ARTmax	Maximum Ambrosio's relational thickness
D-index	Belin/Ambrosio deviation index
AL	Axial length

Supplementary Information

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Supplementary Material 1

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Author contributions

Conceptualization, L.H. and Y.L.; methodology, L.H.; software, L.H.; validation, L.H.; formal analysis, L.H.; investigation, L.H. and Y.L.; resources, Y.Y. and Y.L.; data curation, Y.Y. and X.M.; writing—original draft preparation, L.H.; writing—review and editing, L.H. and Y.Y.; visualization, S.T.; supervision, S.T.; project administration, S.T. and X.M.; funding acquisition, L.H. All authors have read and agreed to the published version of the manuscript.

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Data availability

The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board (or Ethics Committee) of the Ethics Committee of The First Affiliated Hospital of Fujian Medical University (MRTCA, ECFAH of FMU [2022] 323-2). Informed consent was obtained from all the subjects involved in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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